ClearOne Advantage

DAMN!!
(Divya, Ankit, Manasa, Natalie)

Who is our sponsor?

Nathan Broslawsky

Chief Product & Technology Officer
ClearOne Advantage

ClearOne Advantage®

- Debt relief company — helps clients settle credit card & medical debt
- Clients: avg. 52–53 yrs old, $10k–$80k in debt, 8–9 accounts
- Pitch: stop paying cards → build savings → ClearOne negotiates settlements
- Hundreds of agents trained annually on inbound/outbound calls

What Problem Are We Solving?

New agents efficiency struggle

No practice space

Feedback is subjective

Natalie

Personas & Scoring

- Build and maintain customer personas in JSON
- Each persona has financial profile, emotional state, communication style, and dynamic emotion rules
- Personas react differently based on how the agent speaks

- After each session, a second GPT-4o call scores the agent on 7 criteria (Empathy, Clarity, Objection Handling, Active Listening, Compliance, Pacing, Closing)
- Output: plain text report with scores, strengths, improvements, and session summary

3.5/5

Tech Stack and Architecture

Personas: GPT-4o

Scoring: GPT-4o

Persona Storage: JSON files

Conversation Loop: Python

Transcript → Persona: python-docx + GPT-4o

Persona JSON

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System Prompt Builder

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GPT-4o ↔ Conversation History ↔ Agent Input

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Session Transcript

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Evaluator Prompt (GPT-4o, separate call)

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Scored Report (.txt)

Ankit

Voice Implementation

Deepgram -> STT (Speech To Text)

ElevenLabs -> TTS (Text To Speech)

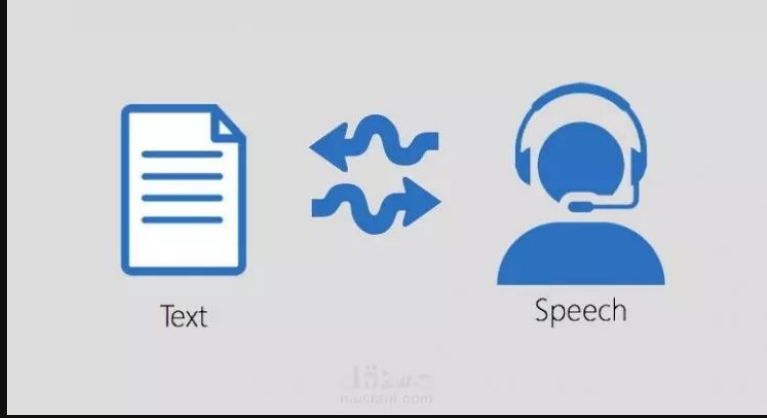
Twilio -> Link between STT, LLM and TTS

Ankit

Voice Implementation

Models used:

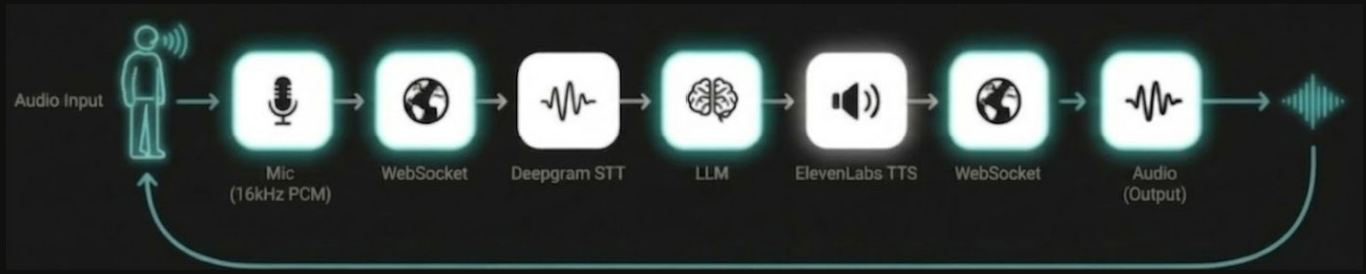
Deepgram (STT)	Nova-2 model – optimized for conversational & financial English
GPT-4o-mini (LLM)	Lower latency than GPT-4o, enough quality to real-time conversation feel
ElevenLabs (TTS)	eleven_flash_v2_5 – speed-optimized, fastest model in their lineup



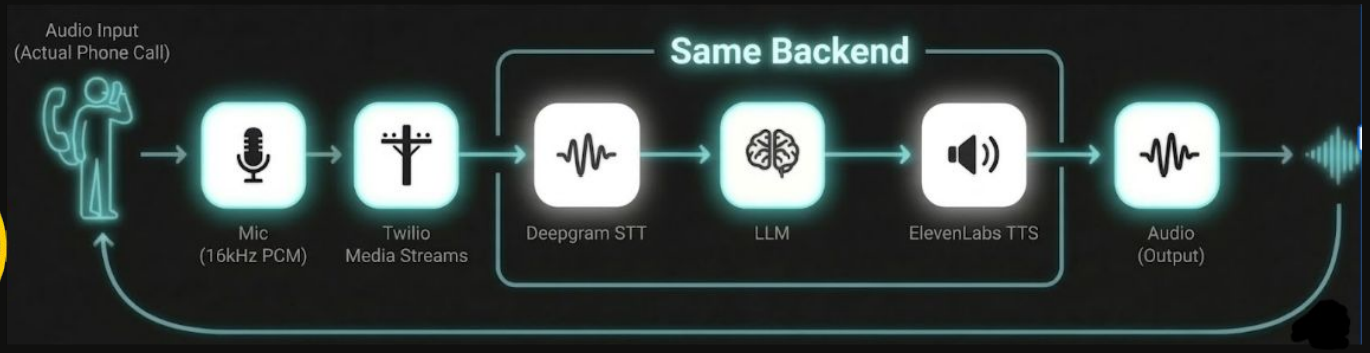
Ankit

Voice Implementation

Browser Path (Currently Demo ready!)



Twilio Path (Actual Planned Implementation)



Manasa

The Interface

- Used React to create a multi-screen application
  - Home, Live Call, Post-Call Report
- Home screen randomly assigns a customer persona to begin each session
- Live screen displays a real-time transcript between the agent and simulated customer
  - UI Integrates Twilio, Deepgram, and ElevenLabs
- Post-call report displays scores based on predetermined categories and classifications
- Annotated transcript screen allows agent to revisit the call line by line, view AI-generated feedback, and add personal notes

Call Report & Feedback

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Session Report — Skeptical Sarah

1.3 / 10

Scorecard | Annotate Transcript

Rapport Building	2/10
Objection Handling	1/10
Clarity	1/10
Empathy	1/10
Next Steps	1/10
Compliance	1/10

✓ STRENGTHS

△ NEEDS IMPROVEMENT

The agent should have introduced themselves and the company clearly at the start of the call to build rapport and trust.

The agent missed an opportunity to acknowledge Sarah's skepticism and reassure her by explaining the company's credibility and process.

← Start New Session

Live Call

LIVE SESSION · 00:30

Skeptical Sarah
Distrustful consumer who has been burned by scams before

SKEPTICAL SARAH

YOU

TRANSCRIPT

SKEPTICAL SARAH
Hello? This is Sarah. Who's calling?

YOU
Hello?

SKEPTICAL SARAH
Hi, I can hear you now. Who's this?

End Call

Review & Annotation

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Session Report — Skeptical Sarah

1.3 / 10

Scorecard | Annotate Transcript

Click any message to see AI feedback · Add your own notes with the comment icon

SKEPTICAL SARAH
Hello? This is Sarah. Who's calling?

YOU
Hello?

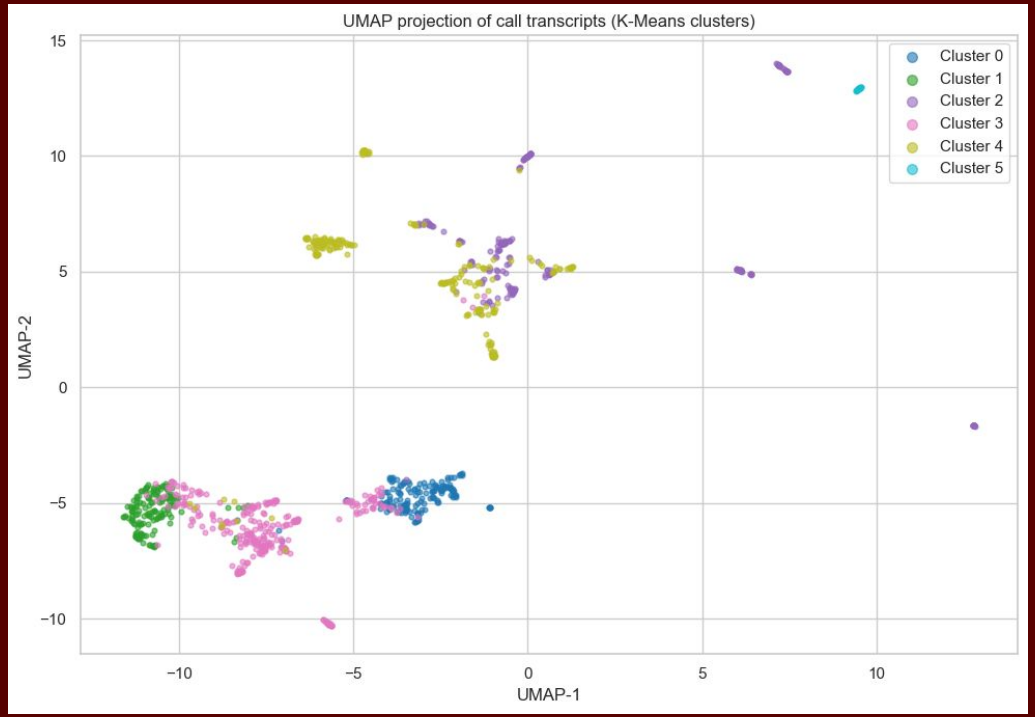
△ IMPROVE
The agent failed to respond appropriately to Sarah's initial inquiry, missing a chance to establish a connection.

SKEPTICAL SARAH
Hi, I can hear you now. Who's this?

← Start New Session

Divya

The Data



- Raw transcript text

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- TF-IDF (5,000 features) — word/bigram frequency matrix

↓

- TruncatedSVD/LSA (100 dimensions) — compresses sparse TF-IDF into dense vectors

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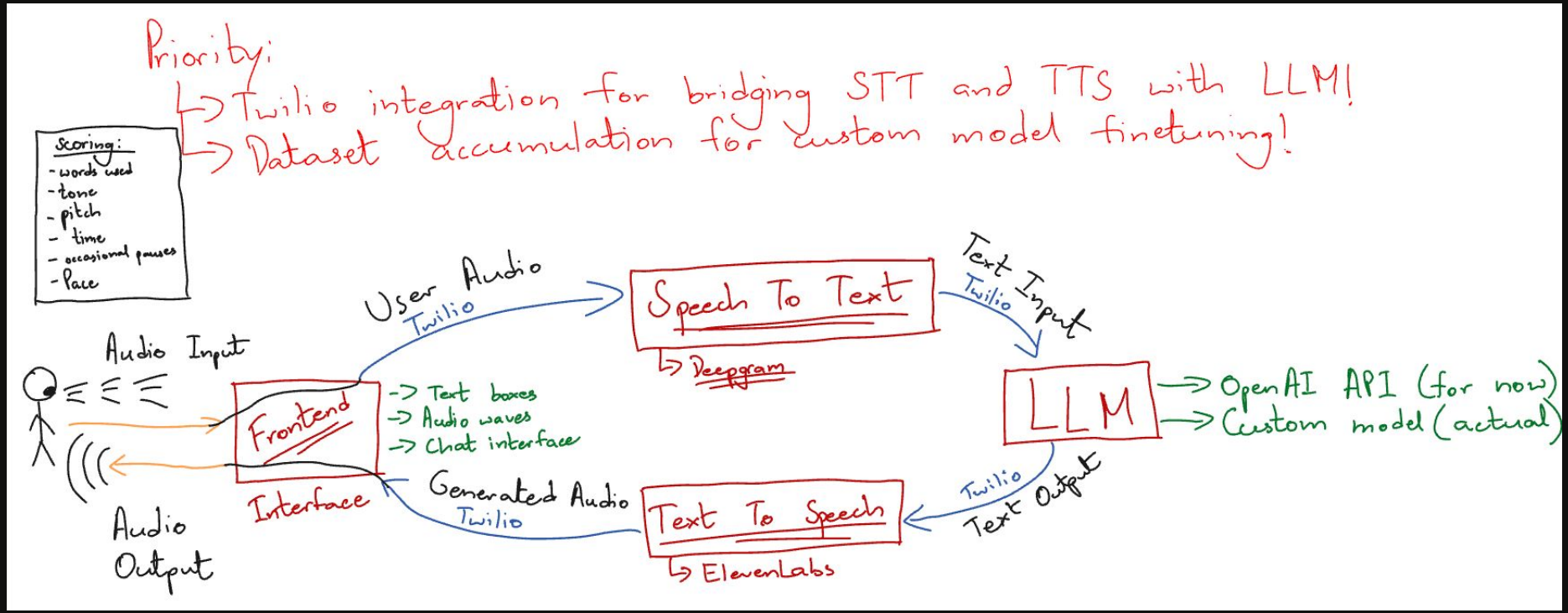
- UMAP (2 dimensions) — preserves neighborhood structure from 100D → 2D

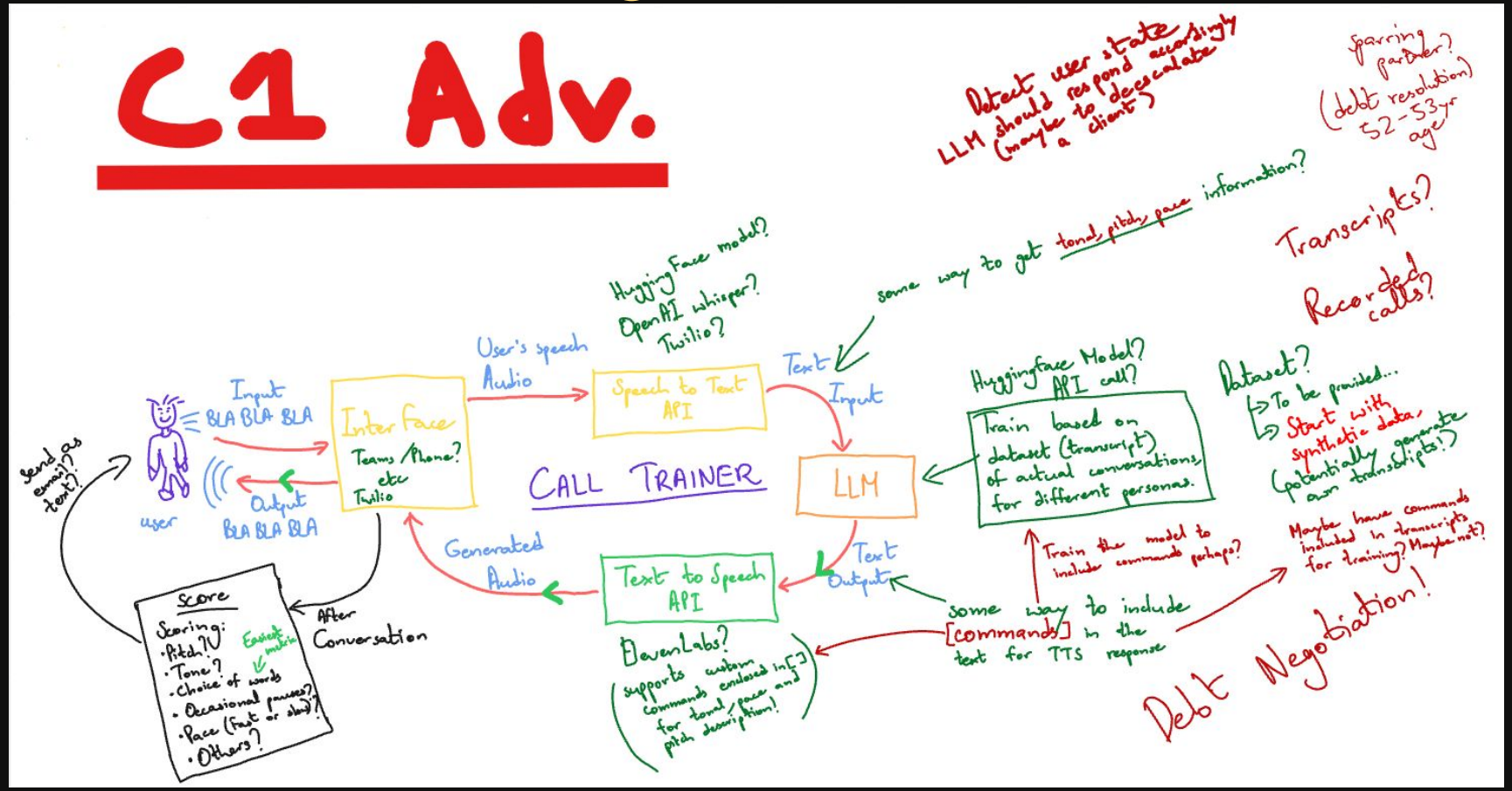
↓

- UMAP-1, UMAP-2

The Architecture



Detailed Planning for Reference



What Is the Hard Part?

1

Realistic Personas

Making the AI customer behave like a real, emotionally complex person, not a chatbot, requires fine-tuned prompts, dynamic emotion rules, and real call transcript data.

2

Consistent Scoring system

Getting LLM-as-judge to score calls consistently and fairly across different conversation styles, without bias or hallucination, is a non-trivial evaluation challenge.

3

Real-time Voice Pipeline

Chaining STT → LLM → TTS with low enough latency to feel like a real phone call requires careful orchestration across Deepgram, GPT-4o-mini, and ElevenLabs.

4

Speech Fragmentation

Small parts of speech were getting sent as different broken prompts, which had to be fixed by storing the different utterances of speech together as parts, and the LLM call would be fired only after the end of speech utterance, detected if there is 1500 milliseconds of sustained silence.

What Is the Interesting Part?

The persona system uses dynamic emotion rules, i.e., if an agent rushes or uses jargon, the client gets angrier. If they validate concerns, the client softens. The AI reacts like a person.

The transcript-to-persona pipeline can turn any real ClearOne call into a new training character, automatically extracting objections, emotional state, and communication style.

A second GPT-4o instance acts as the evaluator, watching the conversation and scoring it against ClearOne's rubric. The system teaches agents by using AI to watch AI.

Thank you!